

The Machine

By

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Introduction and Thanks

The following paper is essentially a more detailed retelling of the previous paper entitled 'Valkyrie' that has been languishing in the file uploads area of the Yahoo Kryptos Group since October 2013. The presence of the paper in the file uploads wasn't widely telegraphed as it was a place holder, summarizing discoveries to that date that fell well short of any meaningful conclusions. That paper also omitted any mention of the holistic nature of the sculpture itself. In the intervening time we have had another clue revealed by Jim Sanborn, which is the word CLOCK for positions 70 through to 74.

As usual the readership is aimed primarily at those in the Kryptos K4 community who are already conversant with the cipher and plain texts of sections K1-K3 as well as the associated tools and techniques for the relevant decryptions. This is a lengthy paper, but no apologies are given for that other than for it being perhaps too brief in places. I have endeavoured to include enough detail to make the reconstruction of the results as clear as possible. Since the initial draft I've edited quite brutally in places in order to add focus and additional clarity to the core concept. There will no doubt be some mistakes and some areas where I could have expanded more with clearer examples and diagrams, but the general premise presented herein is sound and self-consistent, so please feel free to copy any material in this paper in order to illustrate any corrections as part of your own work. I'm happy for this work to be redistributed in its entirety or drawn upon in parts, but please include a citation. Should you wish to make a physical reconstructions of the machine outlined in this paper then remember that all the intellectual property rights for the design remain with Jim Sanborn.

As with sections K1, K2 and K3 anyone wishing to repeat these results for themselves needs only a basic knowledge of cryptography, a pen and some paper.

This introduction would be incomplete without a sincere expression of gratitude to Jim Sanborn, Ed Scheidt, their patrons and any other unnamed parties behind the scenes for seeding the world with such a constructive and educational curiosity.

Lastly I'd like to thank Elonka Dunin who took the time out to cast an eye over my initial paper back in 2010. Anyone unfamiliar with Kryptos that wants additional background may also want to visit her Kryptos site at <http://www.elonka.com/kryptos> before proceeding any further with this document.

Abstract

A number of what are thought to be novel discoveries are hereby put into the public domain for wider discussion and verification. The core finding is that the Kryptos cipher text is layered on top of a numerical surface that is independent of the lettering itself. This surface represents a hardware model on which to perform an analogue computation of the moons cycles in relation to the solar day and year. As a result of this it is conjectured that the sculpture is in fact a symbolic representation of the Antikythera mechanism. It will be shown that the sculpture is capable of recording time at the level of seconds, minutes, hours, days and weeks. It can further be used to capture the calendar, sidereal and synodic months. The largest time span it records is the 76 year Callippic cycle, which is composed of 4 Metonic cycles each of 19 years in length. It is also shown that the two decryption tableau panels can be used to record subsequent Saros cycles and accurately keep track of the exeligmos between eclipses once their initial location and date is identified. In presenting this hypothesis there is a reappraisal of the context in which K3 should be decrypted and indications are given as to which avenues might be explored further to fully decrypt K4 relating to data manipulation and the recovery of the correct algorithmic process for letter selection. Though there are many new insights in this paper the most important is that of putting a new perspective on the sculpture in that it is far more than a redundant copper scroll once all the eventual plaintext is known and reveals instead a new function that takes it into perpetuity.

Additional Background Materials

A good reference guide to the mechanics of the Antikythera mechanism will be necessary to fully appreciate the concepts put forward in this paper. An excellent book on the subject 'A Portable Cosmos', has recently been released by Alexander Jones ISBN 978-0-19-973934-9 and contains many new discoveries and insights many of which were not known or verified in 1990.

Part I

Background

The following information on the background of the Kryptos sculpture was gathered from a range of internet sources.

Kryptos was dedicated on **November the 3rd 1990** on the grounds of the CIA headquarters in Langley Virginia. The main body of the sculpture is a worked copper scroll by the Artist Jim Sanborn who was commissioned by the CIA to create a series of installation pieces for the buildings courtyard. He was additionally loaned the services of a retiring CIA cryptographic expert by the name of Ed Scheidt who was on hand to provide assistance with the cryptographic elements of the underlying text.

Since 1990 there have been three public acknowledgments in relation to the decryption of parts K1, K2 and K3.

A civilian computer scientist named Jim Gillogly went public with a decryption of parts K1, K2 and K3 in 1999 having used computational analysis to arrive at the plain texts.

After this announcement it was revealed that a CIA employee named David Stein was recognised within the CIA as having cracked parts K1, K2 and K3 using a pencil and paper technique in 1998.

It was later announced in 2005 that an NSA team led by Ken Miller, Dennis McDaniels and two other unnamed individuals had deciphered parts K1, K2 & K3 in late 1992 using computational analysis.

In the years since Gillogly's announcement a mystique has continued to grow around the unsolved K4 cipher text. This is partly due to its resistance to yield its final secrets, but also the mass of interest aroused worldwide by Dan Brown's code sleuthing novels and the growing pervasiveness of cryptography in modern life. As a result there has been a mass of internet based collaboration in the public domain ever since, whereby common interest groups and individuals have persistently poured over the clues left in the plain texts of parts K1, K2 and K3 in search of possible breaks.

For anyone not familiar with the layout of the sculptures original cipher text or the plain texts of K1, K2 and K3 then they read as follows.

The Sculpture Text

The text as it appears on the sculpture has been lifted wholesale from Elonka's website (www.elonka.com/kryptos) and is presented in full below.

Complete Cipher Text and Decryption Tableau as it appears on the sculpture.

EMUFPHZLRFAXYUSDJKZLDKRNHGNFIVJ YQTQUXQBQVYUUVLLTREVJYQTMKYRDMFD VFPJUDEEHZWETZYVGWHKKQETGFQJNCE GGWHKK?DQMC PFQZDQMMIAGPFXHQRLG TIMVMZJANQLVKQEDAGDVFRPJUNGEUNA QZGZLECGYUXUEENJTBILBQCRTBJDFHRR YIZETKZEMVDUFKSJHKFWHKUWQLSZFTI HHDDDUVH?DWKBFUFPWNTDFIYCUQZERE EVLDKFEZMOQQJLTUGSYQPFEUNLAVIDX FLGGTEZ?FKZBSFDQVGOGIPUFXXHHRKF FHQNTGPUAECNUVPDJMQCLQUMUNEDFQ ELZZVRRGKFFVOEEXBDMVVPNFQXEZLGRE DNQFMPNZGLFLPMRJQYALMGNUVPDXVKP DQUMEBEDMHDAFMJGZNUPLGEWJLLAETG	ABCDEFGHIJKLMNOPQRSTUVWXYZABCD AKRYPTOSABCDEFGHIJLMNQUVWXXZKRYP BRYPTOSABCDEFGHIJLMNQUVWXXZKRYPT CYPTOSABCDEFGHIJLMNQUVWXXZKRYPTO DPTOSABCDEFGHIJLMNQUVWXXZKRYPTOS ETOSABCDEFGHIJLMNQUVWXXZKRYPTOSA FOSABCDEFGHIJLMNQUVWXXZKRYPTOSAB GSABCDEFGHIJLMNQUVWXXZKRYPTOSABC HABCDEFGHIJLMNQUVWXXZKRYPTOSABCD IBCDEFGHIJLMNQUVWXXZKRYPTOSABCDE JCDEFGHIJLMNQUVWXXZKRYPTOSABCDEF KDEFGHIJLMNQUVWXXZKRYPTOSABCDEFG LEFGHIJLMNQUVWXXZKRYPTOSABCDEFGH MFGHIJLMNQUVWXXZKRYPTOSABCDEFGHI
ENDYAHROHNLSRHEOCPTEOIBIDYSHNAIA CHTNREYULDSLLSLLNOHSNOSMRWXMNE TPRNGATIHNRRAPESLNNELEBLPIIACAE WMTWNDITEENRAHCTENEUDRETNHAEOE TFOLSEDTIWENHAEIOYTEYQHEENCTAYCR EIFTBRSPAMHHEWENATAMATEGYEERLB TEEFOASFIOTUETUAEOTOARMAEERTNRTI BSEDDNIAAHTTMSTEWPIEROAGRIEWFEB AECTDDHILCEIHSITEGOEAOSDDRYDLORIT RKLMLHAGTDHARDPNEOHMGFMFEUHE ECDMRIPFEIMEHNLSSTTRTVDOHW?OBKR UOXOGHULBSOLIFBBWFLRVQPRNGKSSO TWTQSJSSEKZZWATJKLUDIAWINFBNYP VTTMZFPKWGDKZXTJCDIGKUHUAUEKCAR	NGHIJLMNQUVWXXZKRYPTOSABCDEFGHIJL OHIJLMNQUVWXXZKRYPTOSABCDEFGHIJL PIJLMNQUVWXXZKRYPTOSABCDEFGHIJLM QJLMNQUVWXXZKRYPTOSABCDEFGHIJLMN RLMNQUVWXXZKRYPTOSABCDEFGHIJLMNQ SMNQUVWXXZKRYPTOSABCDEFGHIJLMNQU TNQUVWXXZKRYPTOSABCDEFGHIJLMNQUV UQUVWXXZKRYPTOSABCDEFGHIJLMNQUVW VUVWXXZKRYPTOSABCDEFGHIJLMNQUVWX VVWXXZKRYPTOSABCDEFGHIJLMNQUVWXZ XWXXZKRYPTOSABCDEFGHIJLMNQUVWXZK YXZKRYPTOSABCDEFGHIJLMNQUVWXZKR ZZKRYPTOSABCDEFGHIJLMNQUVWXZKRY ABCDEFGHIJKLMNOPQRSTUVWXYZABCD

K1 is highlighted in Mauve, K2 in Blue, K3 in Green and K4 in Red

Known Plain Text for Sections K1, K2 and K3

K 1 (this was encrypted using the Keyword PALIMPSEST)

BETWEEN SUBTLE SHADING AND THE ABSENCE OF LIGHT LIES THE NUANCE OF IQLUSION

K2 (this was encrypted using the Keyword ABSCISSA) Note: This was publicly corrected by Jim Sanborn on April 19th, 2006 who intervened to point out the IDBYROWS should read XLAYERTWO

IT WAS TOTALLY INVISIBLE HOWS THAT POSSIBLE? THEY USED THE EARTHS MAGNETIC FIELD X THE INFORMATION WAS GATHERED AND TRANSMITTED UNDERGUUND TO AN UNKNOWN LOCATION X DOES LANGLEY KNOW ABOUT THIS? THEY SHOULD ITS BURIED OUT THERE SOMEWHERE X WHO KNOWS THE EXACT LOCATION? ONLY WW THIS WAS HIS LAST MESSAGE X THIRTY EIGHT DEGREES FIFTY SEVEN MINUTES SIX POINT FIVE SECONDS NORTH SEVENTY SEVEN DEGREES EIGHT MINUTES FORTY FOUR SECONDS WEST X LAYER TWO

K 3 (this section was not encrypted using the Kryptos Tableau, but instead made use of a 192 character skip sequence to encrypt the plain text)

SLOWLY DESPARATLY SLOWLY THE REMAINS OF PASSAGE DEBRIS THAT ENCUMBERED THE LOWER PART OF THE DOORWAY WAS REMOVED WITH TREMBLING HANDS I MADE A TINY BREACH IN THE UPPER LEFT HAND CORNER AND THEN WIDENING THE HOLE A LITTLE I INSERTED THE CANDLE AND PEERED IN THE HOT AIR ESCAPING FROM THE CHAMBER CAUSED THE FLAME TO FLICKER BUT PRESENTLY DETAILS OF THE ROOM WITHIN EMERGED FROM THE MIST X CAN YOU SEE ANYTHING Q

In addition to the sculpture there are also installations around the courtyard with embedded morse code fragments. Once interpreted these read.

SOS, LUCID MEMORY, T IS YOUR POSITION, SHADOW FORCES, VIRTUALLY INVISIBLE, DIGETAL INTERPRETATU, and RQ (which maybe alternatively be YR depending on which way up its read).

Interventions post decryption of K1-K3

Issues with the decryption of K2

With regards the correct decryption of K2, Jim Sanborn the artist that created Kryptos intervened personally on April 19th 2006, contacting the group moderators in order to say, "No, that last part is wrong." Elonka's Kryptos website at www.elonka.com/kryptos relays the following about this incident.

"He also indicated that there was a missing character on the sculpture, probably something that would have resulted in a plaintext "X" before that section. He said that he had thought that with the missing character, the section in question would have come out to be an unintelligible scramble. Instead, he was astounded to see that by sheer chance, the resulting random text had turned out to be apparently intelligible English, "ID BY ROWS", although that was not what was intended."

The Berlin Clue

On the 21st November 2010 Jim Sanborn once again intervened, but in this instance it was to reveal a clue. Sanborn indicated that a six letter string between positions 64 through to 69 of the Kryptos plaintext for K4 would read BERLIN once decrypted.

The Clock Clue

On the 20th November 2014, Jim Sanborn revealed another clue in the form of the word CLOCK being the plaintext for positions 70 through to 74. The fact that 11 letters of the plaintext were now known now massively reduced the odds of being able to find suitable decryption schemes.

Part II

It will become evident throughout each step that the whole of the Kryptos sculpture represents a larger body of work than previously realised and is only eventually seen for what it is by solving multiple smaller puzzles. Some of these steps are more significant than others, but in total they add up to a single conclusion that is both self-consistent and intellectually satisfying. Viewed as a whole it will be seen that the piece seemingly steps us through the history of cryptography itself as a result of the smaller puzzles that we are expected to solve. This is done quite brilliantly and bears testament to the intellect and artistry of both Scheidt, Sanborn and perhaps others. I suspect that a further study of the cipher and plain texts after this paper has been distributed will yield many more encryption methods, some of which may light a way to a decryption path that does not require foreknowledge of the BERLIN and CLOCK clues respectively.

A note on using a grid to display the sculpture graphically

Throughout this paper the sculpture lettering for the left hand panels will be laid out on a grid that is 33 columns by 28 rows as displayed below. Observe the spaces that pad the right hand side and also note the positioning of the right shifted T that sits outside of majority of the grid system in row 23.

E	M	U	F	P	H	Z	L	R	F	A	X	Y	U	S	D	J	K	Z	L	D	K	R	N	S	H	G	N	F	I	V	J	
Y	Q	T	Q	U	X	Q	B	Q	V	Y	U	V	L	L	T	R	E	V	J	Y	Q	T	M	K	Y	R	D	M	F	D		
V	F	P	J	U	D	E	E	H	Z	W	E	T	Z	Y	V	G	W	H	K	K	Q	E	T	G	F	Q	J	N	C	E		
G	G	W	H	K	K	?	D	Q	M	C	P	F	Q	Z	D	Q	M	M	I	A	G	P	F	X	H	Q	R	L	G			
T	I	M	V	M	Z	J	A	N	Q	L	V	K	Q	E	D	A	G	D	V	F	R	P	J	U	N	G	E	U	N	A		
Q	Z	G	Z	L	E	C	G	Y	U	X	U	E	E	N	J	T	B	J	L	B	Q	C	R	T	B	J	D	F	H	R	R	
Y	I	Z	E	T	K	Z	E	M	V	D	U	F	K	S	J	H	K	F	W	H	K	U	W	Q	L	S	Z	F	T	I		
H	H	D	D	D	U	V	H	?	D	W	K	B	F	U	F	P	W	N	T	D	F	I	Y	C	U	Q	Z	E	R	E		
E	V	L	D	K	F	E	Z	M	O	Q	Q	J	L	T	T	U	G	S	Y	Q	P	F	E	U	N	L	A	V	I	D	X	
F	L	G	G	T	E	Z	?	F	K	Z	B	S	F	D	Q	V	G	O	G	I	P	U	F	X	H	H	D	R	K	F		
F	H	Q	N	T	G	P	U	A	E	C	N	U	V	P	D	J	M	Q	C	L	Q	U	M	U	N	E	D	F	Q			
E	L	Z	Z	V	R	R	G	K	F	F	V	O	E	E	X	B	D	M	V	P	N	F	Q	X	E	Z	L	G	R	E		
D	N	Q	F	M	P	N	Z	G	L	F	L	P	M	R	J	Q	Y	A	L	M	G	N	U	V	P	D	X	V	K	P		
D	Q	U	M	E	B	E	D	M	H	D	A	F	M	J	G	Z	N	U	P	L	G	E	W	J	L	L	A	E	T	G		
E	N	D	Y	A	H	R	O	H	N	L	S	R	H	E	O	C	P	T	E	O	I	B	I	D	Y	S	H	N	A	I	A	
C	H	T	N	R	E	Y	U	L	D	S	L	L	S	L	L	N	O	H	S	N	O	S	M	R	W	X	M	N	E			
T	P	R	N	G	A	T	I	H	N	R	A	R	P	E	S	L	N	N	E	L	E	B	L	P	I	I	A	C	A	E		
W	M	T	W	N	D	I	T	E	E	N	R	A	H	C	T	E	N	E	U	D	R	E	T	N	H	A	E	O	E			
T	F	O	L	S	E	D	T	I	W	E	N	H	A	E	I	O	Y	T	E	Y	Q	H	E	E	N	C	T	A	Y	C	R	
E	I	F	T	B	R	S	P	A	M	H	H	E	W	E	N	A	T	A	M	A	T	E	G	Y	E	E	R	L	B			
T	E	E	F	O	A	S	F	I	O	T	U	E	T	U	A	E	O	T	O	A	R	M	A	E	E	R	T	N	R	T	I	
B	S	E	D	D	N	I	A	A	H	T	T	M	S	T	E	W	P	I	E	R	O	A	G	R	I	E	W	F	E	B		
A	E	C	T	D	D	H	I	L	C	E	I	H	S	I	T	E	G	O	E	A	O	S	D	D	R	Y	D	L	O	R	I	T
R	K	L	M	L	E	H	A	G	T	D	H	A	R	D	P	N	E	O	H	M	G	F	M	F	E	U	H	E				
E	C	D	M	R	I	P	F	E	I	M	E	H	N	L	S	S	T	T	R	T	V	D	O	H	W	?	O	B	K	R		
U	O	X	O	G	H	U	L	B	S	O	L	I	F	B	B	W	F	L	R	V	Q	Q	P	R	N	G	K	S	S	O		
T	W	T	Q	S	J	Q	S	S	E	K	Z	Z	W	A	T	J	K	L	U	D	I	A	W	I	N	F	B	N	Y	P		
V	T	T	M	Z	F	P	K	W	G	D	K	Z	X	T	J	C	D	I	G	K	U	H	U	A	U	E	K	C	A	R		

Examination of the plaintext for hints

Like all encryption schemes, K1, K2 and K3 have weaknesses that made them vulnerable to cryptographic analysis and attack. K4 has so far resisted any form of conventional cryptographic attack, which implies the sculpture text for K4 taken at face value is either not a computable quantity or if it is then uses an encryption scheme that leaves no statistical artefacts in its wake that can be exploited by the cryptographer in order to reconstruct the underlying scheme.

The alternative is that we have to manipulate the sculpture text in some way in order to make it a computable surface and that K4 somehow interacts with this, rather than being a distinct cipher in its own right.

The necessary manipulations that we need to make in order to make the sculpture computable are given by a series of clues present in the plain text of K1, K2 and K3.

There are multiple 'lucky breaks' that can be made by analysing different clues in the plaintext, but significantly any of these initial entry points lead to the same eventual conclusion.

Let's examine the plaintext of K1 again in more detail.

BETWEEN SUBTLE SHADING AND THE ABSENCE OF LIGHT LIES THE NUANCE OF IQLUSION

Let's take this clue at face value and look for the nuances between 'SUBTLE SHADING' and the 'ABSENCE OF LIGHT'. To make this more explicit, let's write them one above the other.

SUBTLE SHADING	SUBTLE SHADING
ABSENCE OF LIGHT	ABSENCE OF LIGHT

Removing the word LIGHT from both lines gives.

SUBE SADN	SUBE SADN
ABSENCE OF	ABSENCE OF

Removing the common letters in both rows to show the subtle difference (the nuance) gives.

USD
CEOF

Let's examine the first two rows of both the K1 cipher text and the K1 plain text again in respect of this result.

E	M	U	F	P	H	Z	L	R	F	A	X	Y	U	S	D	J	K	Z	L	D	K	R	N	S	H	G	N	F	I	V	J	
Y	Q	T	Q	U	X	Q	B	Q	V	Y	U	V	L	L	T	R	E	V	J	Y	Q	T	M	K	Y	R	D	M	F	D		

B	E	T	W	E	E	N	S	U	B	T	L	E	S	H	A	D	I	N	G	A	N	D	T	H	E	A	B	S	E	N	C	
E	O	F	L	I	G	H	T	L	I	E	S	T	H	E	N	U	A	N	C	E	O	F	I	Q	L	U	S	I	O	N		

At first this result may look unremarkable or perhaps even coincidental, but by making one significant change we can make a meaningful insight. Before this is shown it is worth reminding ourselves of another clue given in K3 to show that this step is working to instructions rather than a purely creative leap.

THE REMAINS OF PASSAGE DEBRIS THAT ENCUMBERED THE LOWER PART OF THE DOORWAY WAS REMOVED

If the letter U represents the first letter to be paired with the first letter of K4, then this is effectively our starting point or doorway. Any letters prior to this point may be considered debris that are encumbering the lower part of the doorway. Let's extract these letters and pull the remaining text to the left and then place the extracted letters in the void that was vacated on the right (letters that have been moved are shown shaded on the right). Let's repeat this operation for the plain text.

U	S	D	J	K	Z	L	D	K	R	N	S	H	G	N	F	I	V	J	E	M	U	F	P	H	Z	L	R	F	A	X	Y	
Y	Q	T	Q	U	X	Q	B	Q	V	Y	U	V	L	L	T	R	E	V	J	Y	Q	T	M	K	Y	R	D	M	F	D		

S	H	A	D	I	N	G	A	N	D	T	H	E	A	B	S	E	N	C	B	E	T	W	E	E	N	S	U	B	T	L	E	
E	O	F	L	I	G	H	T	L	I	E	S	T	H	E	N	U	A	N	C	E	O	F	I	Q	L	U	S	I	O	N		

This single manipulation of the cipher and plain texts now draws our attention to a new pattern.

U	S	D	J	K	Z	L	D	K	R	N	S	H	G	N	F	I	V	J	E	M	U	F	P	H	Z	L	R	F	A	X	Y	
Y	Q	T	Q	U	X	Q	B	Q	V	Y	U	V	L	L	T	R	E	V	J	Y	Q	T	M	K	Y	R	D	M	F	D		

S	H	A	D	I	N	G	A	N	D	T	H	E	A	B	S	E	N	C	B	E	T	W	E	E	N	S	U	B	T	L	E	
E	O	F	L	I	G	H	T	L	I	E	S	T	H	E	N	U	A	N	C	E	O	F	I	Q	L	U	S	I	O	N		

The repeating skip sequences of 10 and 9 characters can now be seen to terminating at the boundary delineated by the CEOF marker.

The significance of this repeating pattern every 19 characters is remarkable for a number of reasons, which will be touched upon later, but probably of most importance at this juncture

is the observation that the XY string now terminates the line. This is believed to be significant as the X plane is known mathematically as the Abscissa and the Y plane as the Ordinate, so the top right of the sculpture now seems to imply a definite X and Y axis.

The next clue we are given that is complementary to this rearrangement comes from re-using the same clue as before, which is HOT AIR ESCAPING. Let's take a look at the first three lines of the plain and cipher text under this re-arrangement.

U	S	D	J	K	Z	L	D	K	R	N	S	H	G	N	F	I	V	J	E	M	U	F	P	H	Z	L	R	F	A	X	Y	
Y	Q	T	Q	U	X	Q	B	Q	V	Y	U	V	L	L	T	R	E	V	J	Y	Q	T	M	K	Y	R	D	M	F	D		
V	F	P	J	U	D	E	E	H	Z	W	E	T	Z	Y	V	G	W	H	K	K	Q	E	T	G	F	Q	J	N	C	E		

S	H	A	D	I	N	G	A	N	D	T	H	E	A	B	S	E	N	C	B	E	T	W	E	E	N	S	U	B	T	L	E	
E	O	F	L	I	G	H	T	L	I	E	S	T	H	E	N	U	A	N	C	E	O	F	I	Q	L	U	S	I	O	N		
I	T	W	A	S	T	O	T	A	L	L	Y	I	N	V	I	S	I	B	L	E	H	O	W	S	T	H	A	T	P	O		

It can be seen quite clearly that we have two distinct sections of the plain text. One has a repeating series of 19 characters and the other comprises what at first seems to be irregular row lengths. Both sections contain the word HOT in the vertical plane. The direction of each being reversed. Another important point to note here is that each of the strings forming the word HOT in the vertical plane are crossed by the letters EOF forming a cross.

S	H	A	D	I	N	G	A	N	D	T	H	E	A	B	S	E	N	C	B	E	T	W	E	E	N	S	U	B	T	L	E	
E	O	F	L	I	G	H	T	L	I	E	S	T	H	E	N	U	A	N	C	E	O	F	I	Q	L	U	S	I	O	N		
I	T	W	A	S	T	O	T	A	L	L	Y	I	N	V	I	S	I	B	L	E	H	O	W	S	T	H	A	T	P	O		

If this is still seen to be a coincidence then let's do the math with the repeating rows of 19 characters. Two rows of 19 are 38 and three rows are 57, which are the hours and minutes of the first coordinates given in K2 for the location of the Kryptos sculpture.

Let's remind ourselves again of those coordinates.

THIRTY EIGHT DEGREES FIFTY SEVEN MINUTES SIX POINT FIVE SECONDS NORTH SEVENTY SEVEN DEGREES EIGHT MINUTES FORTY FOUR SECONDS WEST

With this in mind can we push the scheme given above any further?

U	S	D	J	K	Z	L	D	K	R	N	S	H	G	N	F	I	V	J	E	M	U	F	P	H	Z	L	R	F	A	X	Y	
Y	Q	T	Q	U	X	Q	B	Q	V	Y	U	V	L	L	T	R	E	V	J	Y	Q	T	M	K	Y	R	D	M	F	D		
V	F	P	J	U	D	E	E	H	Z	W	E	T	Z	Y	V	G	W	H	K	K	Q	E	T	G	F	Q	J	N	C	E		
G	G	W	H	K	K	?	D	Q	M	C	P	F	Q	Z	D	Q	M	M	I	A	G	P	F	X	H	Q	R	L	G			

To complete the first set of coordinates we are obviously looking for a representation of the number 6.5. The fourth line shows us 6 letters before the question mark, so if we were to take another five after the question mark, assuming that it is somehow indicative of a decimal point then we would be left with 7 spare characters as shown below.

U	S	D	J	K	Z	L	D	K	R	N	S	H	G	N	F	I	V	J	E	M	U	F	P	H	Z	L	R	F	A	X	Y	
Y	Q	T	Q	U	X	Q	B	Q	V	Y	U	V	L	L	T	R	E	V	J	Y	Q	T	M	K	Y	R	D	M	F	D		
V	F	P	J	U	D	E	E	H	Z	W	E	T	Z	Y	V	G	W	H	K	K	Q	E	T	G	F	Q	J	N	C	E		
G	G	W	H	K	K	?	D	Q	M	C	P	F	Q	Z	D	Q	M	M	I	A	G	P	F	X	H	Q	R	L	G			

Before explaining the relevance of this assertion, let's quickly re-examine the first four rows of the corresponding plain text.

S	H	A	D	I	N	G	A	N	D	T	H	E	A	B	S	E	N	C	B	E	T	W	E	E	N	S	U	B	T	L	E	
E	O	F	L	I	G	H	T	L	I	E	S	T	H	E	N	U	A	N	C	E	O	F	I	Q	L	U	S	I	O	N		
I	T	W	A	S	T	O	T	A	L	L	Y	I	N	V	I	S	I	B	L	E	H	O	W	S	T	H	A	T	P	O		
S	S	I	B	L	E	?	T	H	E	Y	U	S	E	D	T	H	E	E	A	R	T	H	S	M	A	G	N	E	T			
I	C	F	I	E	L	D	X	T	H	E	I	N	F	O	R	M	A	T	I	O	N	W	A	S	G	A	T	H	E	R		

We can now see that we have four rows that encode all the coordinates of the latitude given in K2. So far we have only made one alteration, which was to move 13 letters from the beginning of row one to the end. If we look in the vertical plane we can see that the string HOT has been extended in both instances to encompass an anagram of the words SOUTH and NORTH that is only deficient in an U and R respectively.

An important question to ask at each step is whether we have any clues given in the plain text that support such a scheme. In this instance there is a strong indication given that we're on the right track given by the following clues given in the K2 plain text.

IT WAS TOTALLY INVISIBLE HOWS THAT POSSIBLE? THEY USED THE EARTHS MAGNETIC FIELD.

Two aspects of this clue are important, the first is that EARTHSMAGNETICFIELD is a string of 19 characters and secondly that to make a scheme such as the one above INVISIBLE requires the use of a cryptographic technique called Steganography.

Part III

Steganography

Before embarking on this next section we need to clarify an important point. The grid system that the Kryptos letters have been put into is an abstraction, just as geometry is an abstraction of the real world. The individual blocks making the grid have a number value just like coordinates on an X and Y axis. To give an example, three grid blocks means the number three irrespective of the letters that may be in each box. I make this point as we have already made a manipulation of the text, but this has no bearing on the steganography that has been used in Kryptos, so for clarity I shall show the number surface using the original layout rather than with the re-arrangement of the first row used in the previous section.

This abstraction is a powerful one as we shall see that the numerical values associated with the grid system can be thought of as an analogue of the modern day hardware model. The letters themselves can be seen as being data that fill the registers of this hardware and our decryption technique can be thought of as an algorithm that is run using the underlying mechanics of this machine. This may sound confusing at first given that Kryptos is purely a series of letters printed on copper, but once explained, the underlying genius of the sculpture layout will become evident.

Let's fill in what we currently have in terms of embedded coordinates from the last section.

E	M	U	F	P	H	Z	L	R	F	A	X	Y	U	S	D	J	K	Z	L	D	K	R	N	S	H	G	N	F	I	V	J		
Y	Q	T	Q	U	X	Q	B	Q	V	Y	U	V	L	L	T	R	E	V	J	Y	Q	T	M	K	Y	R	D	M	F	D			
V	F	P	J	U	D	E	E	H	Z	W	E	T	Z	Y	V	G	W	H	K	K	Q	E	T	G	F	Q	J	N	C	E			
G	G	W	H	K	K	?	D	Q	M	C	P	F	Q	Z	D	Q	M	M	I	A	G	P	F	X	H	Q	R	L	G				

The number of boxes in rows one and two give us 38. The addition of the next row gives us 57. The next row is 6 followed by a question mark and then 5. So using the exact same technique we now want to find the second set of coordinates for the longitude, which are 77, 8 and 44 respectively. Here we have our block of 77 and our block of 8 terminated by the question mark. The blank rows above the row of 8 and to the left of the 77 number 36 in total.

E	M	U	F	P	H	Z	L	R	F	A	X	Y	U	S	D	J	K	Z	L	D	K	R	N	S	H	G	N	F	I	V	J	
Y	Q	T	Q	U	X	Q	B	Q	V	Y	U	V	L	L	T	R	E	V	J	Y	Q	T	M	K	Y	R	D	M	F	D		
V	F	P	J	U	D	E	E	H	Z	W	E	T	Z	Y	V	G	W	H	K	K	Q	E	T	G	F	Q	J	N	C	E		
G	G	W	H	K	K	?	D	Q	M	C	P	F	Q	Z	D	Q	M	M	I	A	G	P	F	X	H	Q	R	L	G			
T	I	M	V	M	Z	J	A	N	Q	L	V	K	Q	E	D	A	G	D	V	F	R	P	J	U	N	G	E	U	N	A		
Q	Z	G	Z	L	E	C	G	Y	U	X	U	E	E	N	J	T	B	J	L	B	Q	C	R	T	B	J	D	F	H	R	R	
Y	I	Z	E	T	K	Z	E	M	V	D	U	F	K	S	J	H	K	F	W	H	K	U	W	Q	L	S	Z	F	T	I		
H	H	D	D	D	U	V	H	?	D	W	K	B	F	U	F	P	W	N	T	D	F	I	Y	C	U	Q	Z	E	R	E		
E	V	L	D	K	F	E	Z	M	O	Q	Q	J	L	T	T	U	G	S	Y	Q	P	F	E	U	N	L	A	V	I	D	X	
F	L	G	G	T	E	Z	?	F	K	Z	B	S	F	D	Q	V	G	O	G	I	P	U	F	X	H	H	D	R	K	F		
F	H	Q	N	T	G	P	U	A	E	C	N	U	V	P	D	J	M	Q	C	L	Q	U	M	U	N	E	D	F	Q			
E	L	Z	Z	V	R	R	G	K	F	F	V	O	E	E	X	B	D	M	V	P	N	F	Q	X	E	Z	L	G	R	E		
D	N	Q	F	M	P	N	Z	G	L	F	L	F	M	R	J	Q	Y	A	L	M	G	N	U	V	P	D	X	V	K	P		
D	O	U	M	E	B	E	D	M	H	D	A	F	M	J	G	Z	N	U	P	L	G	E	W	J	L	L	A	E	T	G		

By using the same method we used earlier (of addition to the existing coordinates) then we arrive at our required 44 blocks by adding the 8. ***Therefore we now see that the entire set of coordinates given in K2 are embedded completely within the top left of the sculptures letter layout irrespective of the letters themselves.***

E	M	U	F	P	H	Z	L	R	F	A	X	Y	U	S	D	J	K	Z	L	D	K	R	N	S	H	G	N	F	I	V	J		
Y	Q	T	Q	U	X	Q	B	Q	V	Y	U	V	L	L	T	R	E	V	J	Y	Q	T	M	K	Y	R	D	M	F	D			
V	F	P	J	U	D	E	E	H	Z	W	E	T	Z	Y	V	G	W	H	K	K	Q	E	T	G	F	Q	J	N	C	E			
G	G	W	H	K	K	?	D	Q	M	C	P	F	Q	Z	D	Q	M	M	I	A	G	P	F	X	H	Q	R	L	G				
T	I	M	V	M	Z	J	A	N	Q	L	V	K	Q	E	D	A	G	D	V	F	R	P	J	U	N	G	E	U	N	A			
Q	Z	G	Z	L	E	C	G	Y	U	X	U	E	E	N	J	T	B	J	L	B	Q	C	R	T	B	J	D	F	H	R	R		
Y	I	Z	E	T	K	Z	E	M	V	D	U	F	K	S	J	H	K	F	W	H	K	U	W	Q	L	S	Z	F	T	I			
H	H	D	D	D	U	V	H	?	D	W	K	B	F	U	F	P	W	N	T	D	F	I	Y	C	U	Q	Z	E	R	E			
E	V	L	D	K	F	E	Z	M	O	Q	Q	J	L	T	T	U	G	S	Y	Q	P	F	E	U	N	L	A	V	I	D	X		
F	L	G	G	T	E	Z	?	F	K	Z	B	S	F	D	Q	V	G	O	G	I	P	U	F	X	H	H	D	R	K	F			
F	H	Q	N	T	G	P	U	A	E	C	N	U	V	P	D	J	M	Q	C	L	Q	U	M	U	N	E	D	F	Q				
E	L	Z	Z	V	R	G	K	F	F	V	O	E	E	X	B	D	M	V	P	N	F	Q	X	E	Z	L	G	R	E				
D	N	Q	F	M	P	N	Z	G	L	F	L	P	M	R	J	Q	Y	A	L	M	G	N	U	V	P	D	X	V	K	P			
D	Q	U	M	E	B	E	D	M	H	D	A	F	M	J	G	Z	N	U	P	L	G	E	W	J	L	L	A	E	T	G			

We can see that the line breaks and question marks are significant in some way, so to complete our initial observations of the number patterns lets colourise the remaining relationships. One of the interesting relationships that stands out almost immediately is that a block of four letters is immediately followed by four rows of 12 letters making a block of 48 and 4 or 52 in total.

E	M	U	F	P	H	Z	L	R	F	A	X	Y	U	S	D	J	K	Z	L	D	K	R	N	S	H	G	N	F	I	V	J		
Y	Q	T	Q	U	X	Q	B	Q	V	Y	U	V	L	L	T	R	E	V	J	Y	Q	T	M	K	Y	R	D	M	F	D			
V	F	P	J	U	D	E	E	H	Z	W	E	T	Z	Y	V	G	W	H	K	K	Q	E	T	G	F	Q	J	N	C	E			
G	G	W	H	K	K	?	D	Q	M	C	P	F	Q	Z	D	Q	M	M	I	A	G	P	F	X	H	Q	R	L	G				
T	I	M	V	M	Z	J	A	N	Q	L	V	K	Q	E	D	A	G	D	V	F	R	P	J	U	N	G	E	U	N	A			
Q	Z	G	Z	L	E	C	G	Y	U	X	U	E	E	N	J	T	B	J	L	B	Q	C	R	T	B	J	D	F	H	R	R		
Y	I	Z	E	T	K	Z	E	M	V	D	U	F	K	S	J	H	K	F	W	H	K	U	W	Q	L	S	Z	F	T	I			
H	H	D	D	D	U	V	H	?	D	W	K	B	F	U	F	P	W	N	T	D	F	I	Y	C	U	Q	Z	E	R	E			
E	V	L	D	K	F	E	Z	M	O	Q	Q	J	L	T	T	U	G	S	Y	Q	P	F	E	U	N	L	A	V	I	D	X		
F	L	G	G	T	E	Z	?	F	K	Z	B	S	F	D	Q	V	G	O	G	I	P	U	F	X	H	H	D	R	K	F			
F	H	Q	N	T	G	P	U	A	E	C	N	U	V	P	D	J	M	Q	C	L	Q	U	M	U	N	E	D	F	Q				
E	L	Z	Z	V	R	R	G	K	F	F	V	O	E	E	X	B	D	M	V	P	N	F	Q	X	E	Z	L	G	R	E			
D	N	Q	F	M	P	N	Z	G	L	F	L	P	M	R	J	Q	Y	A	L	M	G	N	U	V	P	D	X	V	K	P			
D	Q	U	M	E	B	E	D	M	H	D	A	F	M	J	G	Z	N	U	P	L	G	E	W	J	L	L	A	E	T	G			

This figure of 52 in proximity to a block of 7 and a row above of 12 is interesting as it immediately makes us think of calendar days and weeks. With this in mind let's now take a look at the right hand side of the letter layout.

E	M	U	F	P	H	Z	L	R	F	A	X	Y	U	S	D	J	K	Z	L	D	K	R	N	S	H	G	N	F	I	V	J		
Y	Q	T	Q	U	X	Q	B	Q	V	Y	U	V	L	L	T	R	E	V	J	Y	Q	T	M	K	Y	R	D	M	F	D			
V	F	P	J	U	D	E	E	H	Z	W	E	T	Z	Y	V	G	W	H	K	K	Q	E	T	G	F	Q	J	N	C	E			
G	G	W	H	K	K	?	D	Q	M	C	P	F	Q	Z	D	Q	M	M	I	A	G	P	F	X	H	Q	R	L	G				
T	I	M	V	M	Z	J	A	N	Q	L	V	K	Q	E	D	A	G	D	V	F	R	P	J	U	N	G	E	U	N	A			
Q	Z	G	Z	L	E	C	G	Y	U	X	U	E	E	N	J	T	B	J	L	B	Q	C	R	T	B	J	D	F	H	R	R		
Y	I	Z	E	T	K	Z	E	M	V	D	U	F	K	S	J	H	K	F	W	H	K	U	W	Q	L	S	Z	F	T	I			
H	H	D	D	D	U	V	H	?	D	W	K	B	F	U	F	P	W	N	T	D	F	I	Y	C	U	Q	Z	E	R	E			
E	V	L	D	K	F	E	Z	M	O	Q	Q	J	L	T	T	U	G	S	Y	Q	P	F	E	U	N	L	A	V	I	D	X		
F	L	G	G	T	E	Z	?	F	K	Z	B	S	F	D	Q	V	G	O	G	I	P	U	F	X	H	H	D	R	K	F			
F	H	Q	N	T	G	P	U	A	E	C	N	U	V	P	D	J	M	Q	C	L	Q	U	M	U	N	E	D	F	Q				
E	L	Z	Z	V	R	R	G	K	F	F	V	O	E	E	X	B	D	M	V	P	N	F	Q	X	E	Z	L	G	R	E			
D	N	Q	F	M	P	N	Z	G	L	F	L	P	M	R	J	Q	Y	A	L	M	G	N	U	V	P	D	X	V	K	P			
D	Q	U	M	E	B	E	D	M	H	D	A	F	M	J	G	Z	N	U	P	L	G	E	W	J	L	L	A	E	T	G			

E	N	D	Y	A	H	R	O	H	N	L	S	R	H	E	O	C	P	T	E	O	I	B	I	D	Y	S	H	N	A	I	A	
C	H	T	N	R	E	Y	U	L	D	S	L	L	S	L	L	N	O	H	S	N	O	S	M	R	W	X	M	N	E			
T	P	R	N	G	A	T	I	H	N	R	A	R	P	E	S	L	N	N	E	L	E	B	L	P	I	I	A	C	A	E		
W	M	T	W	N	D	I	T	E	E	N	R	A	H	C	T	E	N	E	U	D	R	E	T	N	H	A	E	O	E			
T	F	O	L	S	E	D	T	I	W	E	N	H	A	E	I	O	Y	T	E	Y	Q	H	E	E	N	C	T	A	Y	C	R	
E	I	F	T	B	R	S	P	A	M	H	H	E	W	E	N	A	T	A	M	A	T	E	G	Y	E	E	R	L	B			
T	E	E	F	O	A	S	F	I	O	T	U	E	T	U	A	E	O	T	O	A	R	M	A	E	E	E	R	T	N	R	T	I
B	S	E	D	D	N	I	A	A	H	T	T	M	S	T	E	W	P	I	E	R	O	A	G	R	I	E	W	F	E	B		
A	E	C	T	D	D	H	I	L	C	E	I	H	S	I	T	E	G	O	E	A	O	S	D	D	R	Y	D	L	O	R	I	T
R	K	L	M	L	E	H	A	G	T	D	H	A	R	D	P	N	E	O	H	M	G	F	M	F	E	U	H	E				
E	C	D	M	R	I	P	F	E	I	M	E	H	N	L	S	S	T	T	R	T	V	D	O	H	W	?	O	B	K	R		
U	O	X	O	G	H	U	L	B	S	O	L	I	F	B	B	W	F	L	R	V	Q	Q	P	R	N	G	K	S	S	O		
T	W	T	Q	S	J	Q	S	S	E	K	Z	Z	W	A	T	J	K	L	U	D	I	A	W	I	N	F	B	N	Y	P		
V	T	T	M	Z	F	P	K	W	G	D	K	Z	X	T	J	C	D	I	G	K	U	H	U	A	U	E	K	C	A	R		

If it is not immediately obvious then the green block on the right hand side consists of 28 rows of 13 blocks making a total of 364. The rather out of place looking T that sits outside of the grid brings us to a grand total of 365. Let's also not forget that the scheme above does not include the 'missed X' in the 14th row, so when added (every leap year) we have a total of 366 squares present on the right hand half of the sculpture. At this point it also becomes apparent that the spaces are important as there are 28 of them, which shows our first potential connection to something relating to the lunar calendar.

From now on we shall show the spaces with an asterisk in order to complete the rows. I will also show the effect of adding the 'missed X' in this next diagram to show the grid in its 366 square leap year mode.

E	M	U	F	P	H	Z	L	R	F	A	X	Y	U	S	D	J	K	Z	L	D	K	R	N	S	H	G	N	F	I	V	J	
Y	Q	T	Q	U	X	Q	B	Q	V	Y	U	V	L	L	T	R	E	V	J	Y	Q	T	M	K	Y	R	D	M	F	D	*	
V	F	P	J	U	D	E	E	H	Z	W	E	T	Z	Y	V	G	W	H	K	K	Q	E	T	G	F	Q	J	N	C	E	*	
G	G	W	H	K	K	?	D	Q	M	C	P	F	Q	Z	D	Q	M	M	I	A	G	P	F	X	H	Q	R	L	G	*	*	
T	I	M	V	M	Z	J	A	N	Q	L	V	K	Q	E	D	A	G	D	V	F	R	P	J	U	N	G	E	U	N	A	*	
Q	Z	G	Z	L	E	C	G	Y	U	X	U	E	E	N	J	T	B	J	L	B	Q	C	R	T	B	J	D	F	H	R	R	
Y	I	Z	E	T	K	Z	E	M	V	D	U	F	K	S	J	H	K	F	W	H	K	U	W	Q	L	S	Z	F	T	I	*	
H	H	D	D	D	U	V	H	?	D	W	K	B	F	U	F	P	W	N	T	D	F	I	Y	C	U	Q	Z	E	R	E	*	
E	V	L	D	K	F	E	Z	M	O	Q	Q	J	L	T	T	U	G	S	Y	Q	P	F	E	U	N	L	A	V	I	D	X	
F	L	G	G	T	E	Z	?	F	K	Z	B	S	F	D	Q	V	G	O	G	I	P	U	F	X	H	H	D	R	K	F	*	
F	H	Q	N	T	G	P	U	A	E	C	N	U	V	P	D	J	M	Q	C	L	Q	U	M	U	N	E	D	F	Q	*	*	
E	L	Z	Z	V	R	R	G	K	F	F	V	O	E	E	X	B	D	M	V	P	N	F	Q	X	E	Z	L	G	R	E	*	
D	N	Q	F	M	P	N	Z	G	L	F	L	P	M	R	J	Q	Y	A	L	M	G	N	U	V	P	D	X	V	K	P	*	
D	O	U	M	E	B	E	D	M	H	D	A	F	M	J	G	Z	N	U	P	L	G	E	S	W	J	L	L	A	E	T	G	

E	N	D	Y	A	H	R	O	H	N	L	S	R	H	E	O	C	P	T	E	O	I	B	I	D	Y	S	H	N	A	I	A	
C	H	T	N	R	E	Y	U	L	D	S	L	L	S	L	L	N	O	H	S	N	O	S	M	R	W	X	M	N	E	*	*	
T	P	R	N	G	A	T	I	H	N	R	A	R	P	E	S	L	N	N	E	L	E	B	L	P	I	I	A	C	A	E	*	
W	M	T	W	N	D	I	T	E	E	N	R	A	H	C	T	E	N	E	U	D	R	E	T	N	H	A	E	O	E	*	*	
T	F	O	L	S	E	D	T	I	W	E	N	H	A	E	I	O	Y	T	E	Y	Q	H	E	E	N	C	T	A	Y	C	R	
E	I	F	T	B	R	S	P	A	M	H	H	E	W	E	N	A	T	A	M	A	T	E	G	Y	E	E	R	L	B	*	*	
T	E	E	F	O	A	S	F	I	O	T	U	E	T	U	A	E	O	T	O	A	R	M	A	E	E	R	T	N	R	T	I	
B	S	E	D	D	N	I	A	A	H	T	T	M	S	T	E	W	P	I	E	R	O	A	G	R	I	E	W	F	E	B	*	
A	E	C	T	D	D	H	I	L	C	E	I	H	S	I	T	E	G	O	E	A	O	S	D	D	R	Y	D	L	O	R	I	T
R	K	L	M	L	E	H	A	G	T	D	H	A	R	D	P	N	E	O	H	M	G	F	M	F	E	U	H	E	*	*	*	
E	C	D	M	R	I	P	F	E	I	M	E	H	N	L	S	S	T	T	R	T	V	D	O	H	W	?	O	B	K	R	*	
U	O	X	O	G	H	U	L	B	S	O	L	I	F	B	B	W	F	L	R	V	Q	Q	P	R	N	G	K	S	S	O	*	
T	W	T	Q	S	J	U	S	S	E	K	Z	Z	W	A	T	J	K	L	U	D	I	A	W	I	N	F	B	N	Y	P	*	
V	T	T	M	Z	F	P	K	W	G	D	K	Z	X	T	J	C	D	I	G	K	U	H	U	A	U	E	K	C	A	R	*	

If the right hand side of the sculpture represents the year 1990, which was not a leap year then the position of the T that sits outside of the grid represents a date of the 28th October, which interestingly is the date that commemorates the end of the Cuban Missile Crisis. If we use the 366 day scheme (i.e. with the missing X replaced) then we have a date of October 27th. You will see that there are 7 spaces from the T until the end of the grid which represents the 31st December. If we count back to the position of T from the 31st December, but do not include the spaces, then we arrive at a date of November 3rd for the 366 day scheme or November the 2nd for the 365 day scheme. If the reader skipped the introductory section then I'll remind them now that the sculpture was unveiled on the 3rd November 1990 the day after the *previous night's full moon*.

Some additional observations about the calendar section can be made immediately that are of significance. The first is that the extra letter corresponding to the missed X is currently added at a position corresponding to June 23rd one day after the summer solstice.

This can be seen below where the calendar rows have been shaded and numbered below to show the distinct month boundaries starting on January 1st in a non leap year.

1	2	3	4	5	6	7	8	9	10	11	12	13	
14	15	16	17	18	19	20	21	22	23	24	25	26	
27	28	29	30	31	1	2	3	4	5	6	7	8	
9	10	11	12	13	14	15	16	17	18	19	20	21	
22	23	24	25	26	27	28	1	2	3	4	5	6	
7	8	9	10	11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30	31	1	2	
3	4	5	6	7	8	9	10	11	12	13	14	15	
16	17	18	19	20	21	22	23	24	25	26	27	28	
29	30	1	2	3	4	5	6	7	8	9	10	11	
12	13	14	15	16	17	18	19	20	21	22	23	24	
25	26	27	28	29	30	31	1	2	3	4	5	6	
7	8	9	10	11	12	13	14	15	16	17	18	19	
20	21	22	23	24	25	26	27	28	29	30	1	2	
3	4	5	6	7	8	9	10	11	12	13	14	15	
16	17	18	19	20	21	22	23	24	25	26	27	28	
29	30	31	1	2	3	4	5	6	7	8	9	10	
11	12	13	14	15	16	17	18	19	20	21	22	23	
24	25	26	27	28	29	30	31	1	2	3	4	5	
6	7	8	9	10	11	12	13	14	15	16	17	18	
19	20	21	22	23	24	25	26	27	28	29	30	1	
2	3	4	5	6	7	8	9	10	11	12	13	14	
15	16	17	18	19	20	21	22	23	24	25	26	27	28
29	30	31	1	2	3	4	5	6	7	8	9	10	
10	11	12	13	14	15	16	17	18	19	20	21	22	
23	24	25	26	27	28	29	30	1	2	3	4	5	
6	7	8	9	10	11	12	13	14	15	16	17	18	
19	20	21	22	23	24	25	26	27	28	29	30	31	

The addition of the missing X to the plaintext has another surprise in store that takes us ever deeper into understanding how we might be expected to use Kryptos in light of this new knowledge. Remember that the conventional 365 day mode of the sculpture coded for WESTIDBYROWS in the green calendar section of row 14 above. With the additional letter put back the plain text now reads WESTXLAYERTWO. The next clue is something of a creative leap that was found purely by chance, but if any reasoning is to be given in hindsight for this step then it is the initial observation of symmetry between the bottom and top of the sculpture. Imagine then that row 15 represents a reflection of row 14. Look at the correspondence of the letters in the calendar portion of rows 14 and 15.

T	E	S	F	O	R	T	Y	F	O	U	R	S	E	C	O	N	D	S	W	E	S	T	X	L	A	Y	E	R	T	W	O		
S	L	O	W	L	Y	D	E	S	P	A	R	A	T	L	Y	S	L	O	W	L	Y	T	H	E	R	E	M	A	I	N	S		

Let's rearrange the letters in the calendar section to arrive at the best possible match to the row above.

T	E	S	F	O	R	T	Y	F	O	U	R	S	E	C	O	N	D	S	W	E	S	T	X	L	A	Y	E	R	T	W	O		
S	L	O	W	L	Y	D	E	S	P	A	R	A	T	L	Y	S	L	O	W	E	S	T	H	L	A	Y	E	R	M	I	N		

The resulting anagram is WESTHLAYERMIN or to be more explicit WEST H LAYER MIN. The H and Min is obviously an allusion to Hours and Minutes, so with this in mind let's reappraise the coordinates we're given in K2, which we should remember are also the coordinates of the sculpture itself.

Part IV

The Coordinates

The coordinates of the sculpture itself as given in K2 are quite remarkable. The bigger question that will no doubt be asked post this paper will be, which came first, the coordinates or the CIA, by which I mean, is this a coincidence or was the siting of the original building itself chosen on the basis of these coordinates? Either way it should remind us that there is an inherent human quality which seemingly causes us to constantly seek out the underlying beauty of mathematical relationships in nature. This appears to have existed in all cultures throughout all history and remains a fundamental aspect of our higher selves that seems to persist even through the darkest ages of ignorance. As a result I would conjecture that there is an aspect of Kryptos that is designed to be timeless and accessible without a knowledge of English. Such properties would mean that these same aspects of the sculptures message would be retrievable by any future generation suitably advanced.

With this in mind let's return to the discovery made at the end of the last section with regards the relevance of the hours and minutes. If we take the anagram located in row 15 at its word and only concentrate on the hours and minutes of the K2 coordinates then we end up with the following relationship when summing the numbers.

$$38 + 57 + 77 + 8 = 180$$

There are a few interesting things to note about this result other than the most obvious which is the geometrical significance of 180. The first is that we have added together two different metrics of hours and minutes to arrive at an answer which is reasonable to suspect is an expression of angular degrees. The second interesting observation about this scheme relates to the numbers encoded in the rows that were discovered in the previous section. We've already conjectured that the rows and question marks are meaningful in terms of delimiting and defining numbers of significance. This assumption leads to the following observation for rows 8-10 of the top left panel.

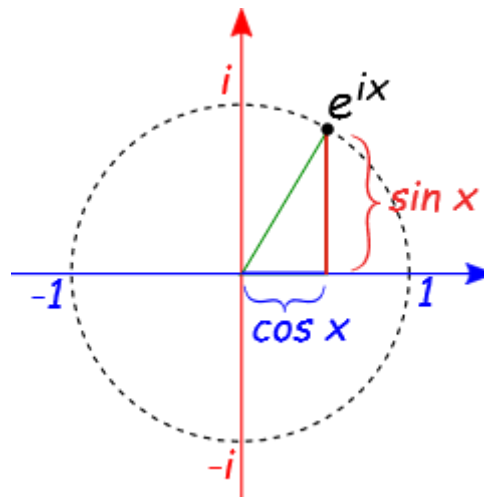
H	H	D	D	D	U	V	H	?	D	W	K	B	F	U	F	P	W	N	T	D	F	I	Y	C	U	Q	Z	E	R	E		
E	V	L	D	K	F	E	Z	M	O	Q	Q	J	L	T	T	U	G	S	Y	Q	P	F	E	U	N	L	A	V	I	D	X	
F	L	G	G	T	E	Z	?	F	K	Z	B	S	F	D	Q	V	G	O	G	I	P	U	F	X	H	H	D	R	K	F		
F	H	Q	N	T	G	P	U	A	E	C	N	U	V	P	D	J	M	Q	C	L	Q	U	M	U	N	E	D	F	Q			

We can see that we have blocks in these rows of 8, 3, 12, 7 and 4 respectively. The blocks of 3, 12, and 7 that are grouped contiguously are of interest as using the same methods of addition that we used previously we can add 7 to 12 to give us 19 and 19 to 3 to give us 22.

On closer inspection we can see that 19 divided by 7 is a close approximation to Euler's number otherwise known as e or 2.718. We can also see another well known ratio of π or 3.14 expressed as 22 and 7. To see e and π in such close proximity immediately conjures up the relations inherent in Euler's Identity. For anyone unfamiliar with this identity it is sometimes referred to as one of the most beautiful equations in mathematics as it forms a link between 5 mathematical fundamentals, which are π , e , 1, 0 and the imaginary unit of i (the square root of -1). The identity is given in full as:

$$e^{i\pi} + 1 = 0$$

If we follow the trail a bit further then we find that the proof of Euler's identity is derived from observations about the interaction between these constants in the real and imaginary geometric planes as shown here:



The allusion to some kind of circular relationship is deemed to be significant, but before delving into this in more detail, let's quickly look at one last relationship thrown up by the numbers participating in the Kryptos coordinates.

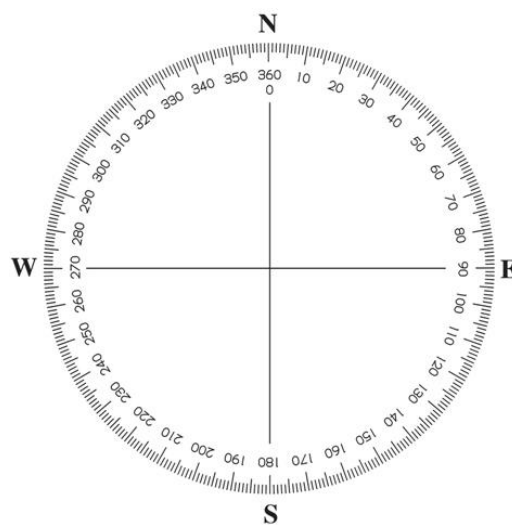
We can see that hours and minutes were considered equivalent in arriving at the previous result, so let's try the same method, but now we'll add the seconds as well. We'll treat each individual component as a discrete number even if it is fractional (we've already summed the hours and minutes to make 180).

$$180 + 6 + 5 + 44 = 235$$

Whilst this number may appear innocuous at first it is absolutely key in taking us down to a deeper level of understanding of what Kryptos is. To be explicit 235 is the number of synodic months in what is known as the Metonic cycle. The Metonic cycle arises as the period of time for an integer relationship to form between the number of solar years and

the number of sidereal and synodic moon months. Something that will be covered later is the number of different ways in which we can measure a moon month, but essentially both the synodic and sidereal months both have integer values in relation to 19 solar years. We have already seen that 19 is an important number in the embedded coordinate scheme, but here we're led one step further towards what appears to be a lunar connection to Kryptos.

With this in mind we can start reappraising other aspects of what we already know. Firstly we know that two complete cycles (or traversal of the embedded coordinate system) of the hours and minutes specified in the embedded coordinate system will denote the traversal of 360 degrees.



On this basis we can make a deductive leap and arrive at a profound result. The result is most likely proof that there is a hardware model in play that is designed to be an integral part of the eventual decryption process.

(A quick note before proceeding is that the coordinate traversal requires revisiting rows 1 and 2 after we have reached 38. They are then traversed a second time before the traversal of row 3 giving a count of 57. I point this out to show that there are indeed 360 squares in the coordinate row traversal).

If for every 180 degree pass through the hours and minutes coordinate rows we accrue an extra 6.5 degrees (as defined symbolically by the last block of 12 characters in the first set of coordinates) then we eventually accrue a total of 13 degrees extra for each 360 traversal.

This is significant as 13 degrees represents the apparent motion of the moon through the sky each day.

If we then use the 28 spaces in the calendar block to record each time that we traverse 13 degrees then we get an averaged way of deriving the 13 lunar months of the year. This time keeping function is interesting, but should not divert our attention away from observing the 'thing' of real significance.

If we choose to either ignore the '?' symbol in the block of 12 characters that make up our 6.5 degree row then every time the cycle repeats we will have skipped forward 192 characters.

This skip sequence will be immediately recognisable as the algorithm required to extract K3.

Therefore all we have to do is set a counter off in K3 that traverses 1 square for every square in the coordinate traversal scheme plus the extra 11 squares representing half a day's worth of lunar traversal. At the point the cycle repeats we should record each letter currently under the K3 traversal counter.

If we are dealing with a machine that is capable of tracking the moons progress in relation to the solar year then there is already a precedent set in this regard by way of a mechanism from antiquity known as the Antikythera mechanism.

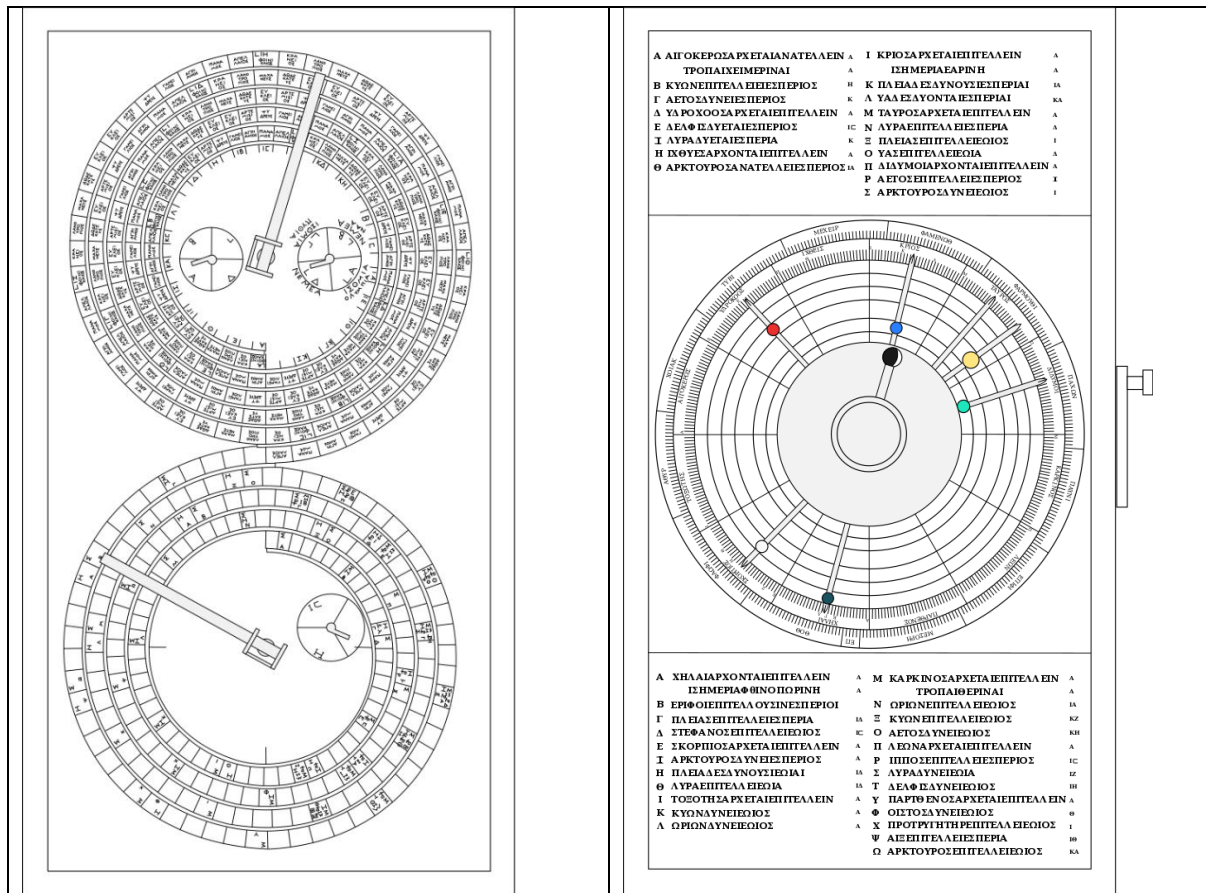
Part V

The Antikythera Mechanism

For those not familiar with the Antikythera Mechanism it is essentially the remains of some heavily corroded metal gears found on a ship wreck off the island of Antikythera at the very beginning of the 20th century. This strange artefact lay unclassified in a Greek museum for some time before archaeologists suddenly noted that the gearing ratios in the mechanism appeared to relate to celestial cycles. As the sophistication of the device became apparent people started to refer to the mechanism as an object seemingly displaced in time as it encapsulated a detailed knowledge of celestial mechanics thought impossible to have been known at the time of construction. In short the mechanism encapsulates the workings of both the Metonic and Saros cycles of 19 and approximately 18 years respectively. These cycles are important as they can be used to predict the days on which solar and lunar eclipses are likely to occur. The aspects of eclipse prediction will be revisited in a later section, but first lets acquaint ourselves with a reconstruction of what the Antikythera Mechanism looked like. A good starting point as ever is the Wikipedia entry at: https://en.wikipedia.org/wiki/Antikythera_mechanism, but we should at least note here that there were three major dials that could be read off from the mechanism. The mechanism had a front and back face, with the front having a single dial and the back having two dials. There was a winding arm in the mechanism that would allow the operator to animate the device and show the evolution of the relative positions of the moon and planets with respect to time. The largest remaining part of the mechanism recovered from the seabed can be seen below. It was only with the advent of advanced imaging techniques that the true complexity of the gearing became apparent.



Once reconstructed, the front and back dials of the mechanism looked as follows. The dials of most interest to us are the ones on the back face of the device as shown here on the left as they represent both the Metonic and Saros cycles respectively.



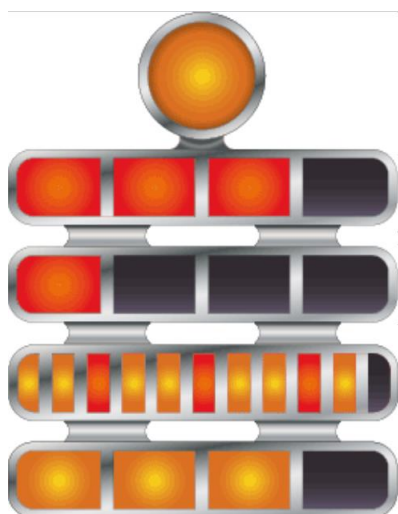
For anyone looking to find out more about the Antikythera mechanism after reading this paper then an excellent resource is the recent book on the subject 'A portable Cosmos' by Alexander Jones, which is a very up to date summation of all the history , research and recent conclusions. One of the most remarkable aspects of the mechanism was that it was covered in something akin to an instruction manual that gave a good indication of how it should be used.

Part VI

Putting it all together

We should not be at all surprised that we are being asked to theoretically reconstruct some kind of machine in order to gain deeper insights into how we might finally break Kryptos. There is an important precedent already set in the history of cryptography in that both the Japanese Purple and German Lorenz machines were first reconstructed theoretically without the cryptanalysts ever having seen one. The knowledge of the machine architectures themselves were crucial in allowing mathematicians to establish how their cryptographic potency might be diminished under certain circumstances, so armed with a long crib such as BERLIN CLOCK it is hoped that this paper will spark a debate about how 'the machine' should be correctly assembled. A first stab at that possible arrangement is now put forward, but before doing so I'll include a brief aside on the mechanics of the Megenlehreuhr clock.

Ever since the release of the clock clue in 2013 there has been speculation with regards the relevance of the Megenlehreuhr clock in Berlin. If we examine the workings of the clock in more detail then we can see it is broken down as follows. In describing the meaning of the layout we can proceed in order of time periods.



The orange light at the top blinks every second.

The set of four lights in the top row represent five hours each.

The four lights in row two represent hours.

The eleven lights in this row represent five minute time periods.

The four lights on the bottom row each represent one minute.

We have already discussed what appears to be a pretty concrete link to the Metonic cycle, but this cycle was considered slightly inaccurate in antiquity so was rounded by taking four Metonic cycles in a row to create a closer match to an integer relationship between solar years and synodic moon months. A period of four Metonic cycles is called the Callippic cycle and represents a period of 76 years. If we think about the scheme of the Megenlehreuhr clock above in relation to this cycle of 76 years then we can conjecture that our first 3 rows of 19 characters in the 38 and 57 coordinate blocks respectively are capable of recording the first 3 Metonic cycles by using a single block for each year. The fourth Metonic cycle could be tracked by tracing out every block in the coordinate scheme whenever a synodic month

completes as we know that this number is 235. Once this number is reached we can empty all the registers and start again from scratch at the beginning of a new Callippic cycle. In the diagram below the likely register positions for other calendar metrics is put forward at the same time.

E	M	U	F	P	H	Z	L	R	F	A	X	Y	U	S	D	J	K	Z	< 19 Years
Y	Q	T	Q	U	X	Q	B	Q	V	Y	U	V	L	L	T	R	E	V	< 38 Years
V	F	P	J	U	D	E	E	H	Z	W	E	T	Z	Y	V	G	W	H	< 57 Years
G	G	W	H	K	K	?	D	Q	M	C	P	F	Q	Z	D	Q	M	M	
T	I	M	V	M	Z	J	A	N	Q	L	V	K	Q	E	D	A	G	D	
Q	Z	G	Z	L	E	C	G	Y	U	X	U	E	E	N	J	T	B	J	
Y	I	Z	E	T	K	Z	E	M	V	D	U	F	K	S	J	H	K	F	
H	H	D	D	D	U	V	H	?	D	W	K	B	F	U	F	P	W	N	
E	V	L	D	K	F	E	Z	M	O	Q	Q	J	L	T	T	U	G	S	< 12 Calendar Months
F	L	G	G	T	E	Z	?	F	K	Z	B	S	F	D	Q	V	G	O	< 7 Days / 52 Weeks
F	H	Q	N	T	G	P	U	A	E	C	N	U	V	P	D	J	M	Q	<
E	L	Z	Z	V	R	R	G	K	F	F	V	O	E	E	X	B	D	M	<
D	N	Q	F	M	P	N	Z	G	L	F	L	P	M	R	J	Q	Y	A	<
D	Q	U	M	E	B	E	D	M	H	D	A	F	M	J	G	Z	N	U	<

It should also be remembered that there are 28 spaces marked with asterisks at the end of the calendar rows. It is conjectured that these are used to record the progress of the averaged moon month of 28 days.

There is obviously a void between rows 4 -8 that doesn't appear to code for anything obvious, so it is asserted that some kind of binary scheme is in play, which may also make the mechanics of the set theory clock relevant to Kryptos. If this theory holds water then is there a logical place for such a scheme to sit within our mechanism? It is in fact possible to create both a binary clock and a binary calendar using 6 bits and 5 bits respectively. We know that we already have a perfect contender for both a 6 and 5 bit header in the form of the seconds from the latitude coordinates. The presence of 3 non spilt rows beneath them also gives us a good place to try and insert such a scheme. First let's take a look at what a binary clock looks like in practice. Such a scheme is shown here and is lifted directly from the wikipedia entry at: https://en.wikipedia.org/wiki/Binary_clock

	Hours	Minutes	Seconds
32		1	1
16	0	0	1
8	1	0	0
4	0	1	0
2	1	0	0
1	0	1	1
	10	37	49

If we rotate this scheme 90 anticlockwise and try placing it first in the mirrored row of the plaintext then we see an interesting correlation. The time above will be displayed in binary where the line corresponding to 1, 2, 4, 8, 16, and 32 is represented as THEMIS. You will notice that the end of the row is bounded by the letters HMA. Could this be a direct pointer towards Hours and Minutes. It also seems to be rather coincidental that rearrangement of THEMIS also gives us TIME HS.

S	L	O	W	L	Y	D	E	S	P	A	R	A	T	L	Y	S	L	O	W	L	Y	T	H	E	R	E	M	A	I	N	S	
O	F	P	A	S	S	A	G	E	D	E	B	R	I	S	T	H	A	T	E	N	C	U	M	B	E	R	E	D	*	*		
H	E	L	O	W	E	R	P	A	R	T	O	F	T	H	E	D	O	O	R	W	A	Y	W	A	S	R	E	M	O	V	*	
E	D	W	I	T	H	T	R	E	M	B	L	I	N	G	H	A	N	D	S	I	M	A	D	E	A	T	I	N	Y	*	*	
B	R	E	A	C	H	I	N	T	H	E	U	P	P	E	R	L	E	F	T	H	A	N	D	C	O	R	N	E	R	A	N	
D	T	H	E	N	W	I	D	E	N	I	N	G	T	H	E	H	O	L	E	A	L	I	T	T	L	E	I	I	N	*	*	
S	E	R	T	E	D	T	H	E	C	A	N	D	L	E	A	N	D	P	E	E	R	E	D	I	N	T	H	E	H	O	T	
A	I	R	E	S	C	A	P	I	N	G	F	R	O	M	T	H	E	C	H	A	M	B	E	R	C	A	U	S	E	D	*	
T	H	E	F	L	A	M	E	T	O	F	L	I	C	K	E	R	B	U	T	P	R	E	S	E	N	T	L	Y	D	E	T	A
I	L	S	O	F	T	H	E	R	O	O	M	W	I	T	H	I	N	E	M	E	R	G	E	D	F	R	O	M	*	*	*	
T	H	E	M	I	S	T	X	C	A	N	Y	O	U	S	E	E	A	N	Y	T	H	I	N	G	Q	?	X	X	X	X	*	
X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	*	
X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	B	E	R	*
L	I	N	C	L	O	C	K	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	*	

Using five bit binary it is also possible to represent the day of the week, calendar month and day of month in the three line slot adjacent to the binary clock. In choosing which row should perform which count we can take some guidance from the mirrored row above, whereby HMA is replaced by DRL. If we use some creative license to imagine that this might be Days, Rotations and Lunations then we can represent Sunday 30th as 7, 30 and 7 as follows.

S	H	A	D	I	N	G	A	N	D	T	H	E	A	B	S	E	N	C	B	E	T	W	E	E	N	S	U	B	T	L	E	
E	O	F	L	I	G	H	T	L	I	E	S	T	H	E	N	U	A	N	C	E	O	F	I	Q	L	U	S	I	O	N	*	
I	T	W	A	S	T	O	T	A	L	L	Y	I	N	V	I	S	I	B	L	E	H	O	W	S	T	H	A	T	P	O	*	
S	S	I	B	L	E	?	T	H	E	Y	U	S	E	D	T	H	E	E	A	R	T	H	S	M	A	G	N	E	T		*	
I	C	F	I	E	L	D	X	T	H	E	I	N	F	O	R	M	A	T	I	O	N	W	A	S	G	A	T	H	E	R	*	
E	D	A	N	D	T	R	A	N	S	M	I	T	T	E	D	U	N	D	E	R	G	R	U	U	N	D	T	O	A	N	U	
N	K	N	O	W	N	L	O	C	A	T	I	O	N	X	D	O	E	S	L	A	N	G	L	E	Y	K	N	O	W	A	*	
B	O	U	T	T	H	I	S	?	T	H	E	Y	S	H	O	U	L	D	I	T	S	B	U	R	I	E	D	O	U	T	*	
T	H	E	R	E	S	O	M	E	W	H	E	R	E	X	W	H	O	K	N	O	W	S	T	H	E	E	X	A	C	T	L	
O	C	A	T	I	O	N	?	O	N	L	Y	W	W	T	H	I	S	W	A	S	H	I	S	L	A	S	T	M	E	S	*	
S	A	G	E	X	T	H	I	R	T	Y	E	I	G	H	T	D	E	G	R	E	E	S	F	I	F	T	Y	S	E	*	*	
V	E	N	M	I	N	U	T	E	S	S	I	X	P	O	I	N	T	F	I	V	E	S	E	C	O	N	D	S	N	O	*	
R	T	H	S	E	V	E	N	T	Y	S	E	V	E	N	D	E	G	R	E	S	E	I	G	H	T	M	I	N	U	*		
T	E	S	F	O	R	T	Y	F	O	U	R	S	E	C	O	N	D	S	W	E	S	T	X	L	A	Y	E	R	T	W	O	*

So if we have a pulse that ticks on a regular basis then we know that we can relate it to the scheme we found earlier for the solar day and the 360 degree coordinate traversal. The only amendment we have to make is that we will only move one coordinate block for every four minutes of the binary clock.

So reappraising the scheme we arrived at before lets have another look at the potential layout for the registers in the machine. It's conjectured that rows 4 - 8 all represent binary quantities including the time and date schemes already discussed. Row four itself is capable of holding a 12 bit number, which would make it an ideal candidate for recording the current year. Row eight has the potential to hold binary counters of 8 and 3 bits respectively, which makes it perfect for recording the 254 sidereal moon months or the 235 synodic moon months in the Metonic cycle. The 3 bit register is interesting as it gives us the possibility of recording a recurring 8 year cycle, which would allow us to know when to switch back and forth between our 365 and 366 letter layout in the calendar rows. This possibility alone is exciting as it implies that there is a dynamism to the sculpture that causes it to pulsate every four years with the addition of an extra letter.

E	M	U	F	P	H	Z	L	R	F	A	X	Y	U	S	D	J	K	Z	< 19 Years
Y	Q	T	Q	U	X	Q	B	Q	V	Y	U	V	L	L	T	R	E	V	< 38 Years
V	F	P	J	U	D	E	E	H	Z	W	E	T	Z	Y	V	G	W	H	< 57 Years
G	G	W	H	K	K	?	D	Q	M	C	P	F	Q	Z	D	Q	M	M	< Binary Years 12 bit
T	I	M	V	M	Z	J	A	N	Q	L	V	K	Q	E	D	A	G	D	< Binary Time/Date
Q	Z	G	Z	L	E	C	G	Y	U	X	U	E	E	N	J	T	B	J	<
Y	I	Z	E	T	K	Z	E	M	V	D	U	F	K	S	J	H	K	F	<
H	H	D	D	D	U	V	H	?	D	W	K	B	F	U	F	P	W	N	< 8 bit / 3 bit register
E	V	L	D	K	F	E	Z	M	O	Q	Q	J	L	T	T	U	G	S	< 12 Calendar Months
F	L	G	G	T	E	Z	?	F	K	Z	B	S	F	D	Q	V	G	O	< 7 Days / 52 Weeks
F	H	Q	N	T	G	P	U	A	E	C	N	U	V	P	D	J	M	Q	<
E	L	Z	Z	V	R	R	G	K	F	F	V	O	E	E	X	B	D	M	<
D	N	Q	F	M	P	N	Z	G	L	F	L	P	M	R	J	Q	Y	A	<
D	Q	U	M	E	B	E	D	M	H	D	A	F	M	J	G	Z	N	U	<

It is of course possible to show the Callippic progression in years in the 77 coordinate block, but there seems to be a greater elegance to using the first 3 rows of 19. In terms of our start point we already know that there was a full moon the Friday night before November 3rd 1990 so we should be able to sync the machine to match the correct celestial representations for any time prior and post this date.

Part VII

Eclipse Prediction

If the machine is indeed capable of recording synodic months then this would imply that it also has the ability to predict eclipses as there is a lunar cycle known as the Saros cycle that repeats every 223 synodic months. The alternative way to calculate this same cycle is to take a period of 18 years and 11 days. If we take the hanging L on the decryption tableau panels as being an analogue to the extraneous T outlined in the previous calendar scheme then we can again construct a 28 row calendar that has 18 preceding characters rather than the other calendars 19.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	
A	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	
B	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	
C	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	
D	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	
E	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	
F	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	
G	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	
H	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	
I	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	
J	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	
K	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	
L	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	
M	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	
N	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L
O	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	
P	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	
Q	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	
R	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	
S	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	
T	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	
U	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	
V	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	
W	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	
X	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	
Y	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	
Z	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	

Why would we need two calendars in one piece of work? Just like the back of the Antikythera mechanism it would appear that we essentially have two dial faces. The first already discussed shows the passing of time with respect to the Metonic and Callippic cycles, but the right hand panels relating to the decryption tableau would appear to relate directly to the Saros cycle, which rather excitingly gives us the ability to create an eclipse calendar based on a phenomenon known as the exeligmos. The exeligmos is the 18 years and 11 days between each saros cycle. There is an additional 8 hour drift westwards of

successive eclipses with each recurrence of the Saros cycle, so this rather round figure of exactly a third of a day can be used to predict with reasonable accuracy the date on which the next eclipse will occur. Because of the 120 degree drift to the west (8 hours is a third of 360) with each Saros it will not be until the third saros is completed that the eclipse will occur again in roughly the same location.

So how can we use the decryption tableau as a calculator. The first thing we have to do is assume that we have some symmetry in play and put back a B in the sixth row that we can add on a leap year.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	
A	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	
B	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	
C	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	
D	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	
E	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B
F	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	
G	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	
H	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	
I	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	
J	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	
K	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	
L	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	
M	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	
N	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L
O	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	
P	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	
Q	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	
R	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	
S	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	
T	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	
U	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	
V	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	
W	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	
X	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	
Y	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	
Z	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	

You will notice that the hanging L is in the mirror row to where the missing X was added. It's by this same symmetry argument that we reserve a B in row six for our leap year arrangement when needed.

The next thing you need to do is find the date of an eclipse that is or was visible from your location. A good example is to use August the 21st 2017 as we know we have a prominent eclipse on this date. If we then set the clock counters to zero and then let them run then we can fill in a square on the left hand side every time a year has passed.

Once we have accumulated 18 years on the left hand side of the row with the last eclipse marker then we can go to step two of the algorithm. As soon as this operation completes then push the date marker along eleven times each time the clock records one day.

If the date marker has moved to a new row in the calendar then repeat this process again using the corresponding row otherwise stay in the same row. Light up one of the notches in the corners each time this process is run. If we have the two notches already lit up, then we know that the next run will represent a full exeligmos and that the eclipse will occur again in roughly the same location on this date. This process works for both solar and lunar eclipses.

Rather intriguingly the pattern formed by the shift in dates every cycle leads to a regular pattern that is only broken in the rows with protruding characters. Let's highlight below what the 11 day shift in dates between each Saros looks like when represented on this calendar.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D		
A	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P		
B	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T		
C	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O		
D	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S		
E	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A		
F	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B		
G	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C		
H	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D		
I	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E		
J	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F		
K	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G		
L	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H		
M	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I		
N	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	
O	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L		
P	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M		
Q	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N		
R	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q		
S	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U		
T	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V		
U	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W		
V	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X		
W	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z		
X	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K		
Y	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R		
Z	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y		
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D		

If we look at the initial pattern we first observed in part one for the first four letters of K4 coding for UNYY to create Kryp when encrypted but look at the corresponding rows in the plaintext instead of the cipher text then we see the following intriguing relationship.

S	H	A	D	I	N	G	A	N	D	T	H	E	A	B	S	E	N	C	B	E	T	W	E	E	N	S	U	B	T	L	E	
E	O	F	L	I	G	H	T	L	I	E	S	T	H	E	N	U	A	N	C	E	O	F	I	Q	L	U	S	I	O	N	*	
I	T	W	A	S	T	O	T	A	L	L	Y	I	N	V	I	S	I	B	L	E	H	O	W	S	T	H	A	T	P	O	*	
S	S	I	B	L	E	?	T	H	E	Y	U	S	E	D	T	H	E	E	A	R	T	H	S	M	A	G	N	E	T		*	
I	C	F	I	E	L	D	X	T	H	E	I	N	F	O	R	M	A	T	I	O	N	W	A	S	G	A	T	H	E	R	*	
E	D	A	N	D	T	R	A	N	S	M	I	T	T	E	D	U	N	D	E	R	G	R	U	U	N	D	T	O	A	N	U	
N	K	N	O	W	N	L	O	C	A	T	I	O	N	X	D	O	E	S	L	A	N	G	L	E	Y	K	N	O	W	A	*	
B	O	U	T	T	H	I	S	?	T	H	E	Y	S	H	O	U	L	D	I	T	S	B	U	R	I	E	D	O	U	T	*	
T	H	E	R	E	S	O	M	E	W	H	E	R	E	X	W	H	O	K	N	O	W	S	T	H	E	E	X	A	C	T	L	
O	C	A	T	I	O	N	?	O	N	L	Y	W	W	T	H	I	S	W	A	S	H	I	S	L	A	S	T	M	E	S	*	
S	A	G	E	X	T	H	I	R	T	Y	E	I	G	H	T	D	E	G	R	E	E	S	F	I	F	T	Y	S	E	*	*	
V	E	N	M	I	N	U	T	E	S	S	I	X	P	O	I	N	T	F	I	V	E	S	E	C	O	N	D	S	N	O	*	
R	T	H	S	E	V	E	N	T	Y	S	E	V	E	N	D	E	G	R	E	E	S	E	I	G	H	T	M	I	N	U	*	
T	E	S	F	O	R	T	Y	F	O	U	R	S	E	C	O	N	D	S	W	E	S	T	X	L	A	Y	E	R	T	W	O	*
S	L	O	W	L	Y	D	E	S	P	A	R	A	T	L	Y	S	L	O	W	L	Y	T	H	E	R	E	M	A	I	N	S	
O	F	P	A	S	S	A	G	E	D	E	B	R	I	S	T	H	A	T	E	N	C	U	M	B	E	R	E	D	T	*	*	
H	E	L	O	W	E	R	P	A	R	T	O	F	T	H	E	D	O	O	R	W	A	Y	W	A	S	R	E	M	O	V	*	
E	D	W	I	T	H	T	R	E	M	B	L	I	N	G	H	A	N	D	S	I	M	A	D	E	A	T	I	N	Y	*	*	
B	R	E	A	C	H	I	N	T	H	E	U	P	P	E	R	L	E	F	T	H	A	N	D	C	O	R	N	E	R	A	N	
D	T	H	E	N	W	I	D	E	N	I	N	G	T	H	E	H	O	L	E	A	L	I	T	T	L	E	I	I	N	*	*	
S	E	R	T	E	D	T	H	E	C	A	N	D	L	E	A	N	D	P	E	E	R	E	D	I	N	T	H	E	H	O	T	
A	I	R	E	S	C	A	P	I	N	G	F	R	O	M	T	H	E	C	H	A	M	B	E	R	C	A	U	S	E	D	*	
T	H	E	F	L	A	M	E	T	O	F	L	I	C	K	E	R	B	U	T	P	R	E	S	E	N	T	L	Y	D	E	T	A
I	L	S	O	F	T	H	E	R	O	O	M	W	I	T	H	I	N	E	M	E	R	G	E	D	F	R	O	M	*	*	*	
T	H	E	M	I	S	T	X	C	A	N	Y	O	U	S	E	E	A	N	Y	T	H	I	N	G	Q	?	X	X	X	X	*	
X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	*	
X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	B	E	R	*
L	I	N	C	L	O	C	K	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	*	

Whether this is coincidence or not remains to be seen, but I would conjecture that the calculation of eclipse days is most likely significant in relation to the eventual decryption of K4. I have not dwelt on any speculative mechanisms for decrypting K4 in this paper other than the proposed machine based revision of the K3 decryption process and the first four letters. Whether the 40 character calendar portion of K4 is distinct from the 57 character block may also be worthy of future debate.

Part IX

Conclusions

We've seen how the machine once reconstructed works as a timepiece capable of recording time, days, months and years over a 76 year period. We can also see that the panels making up the decryption tableau can be used to record subsequent Saros cycles and by extension calculate the date of the exeligmos with reasonable accuracy. Whether or not there is an analogue corresponding to the third face of the Antikythera mechanism remains to be seen, but the inclusion of the anagrams of HYDRA (a constellation in the plane of the ecliptic), SOL and DIANA (the Roman name for the moon goddess) in K3 implies that there may well be an as yet undiscovered analogue to the front face of the Antikythera mechanism that has not been covered in this paper.

Something that hasn't escaped my attention and that I've spent many walks pondering is another aspect of 'The Machine' that Kryptos forces us to contend with. In 1990 Quantum Cryptography would have been at the very cutting edge of computational and mathematical research, whilst today we take it for granted as a reality of our ever increasing range of technological capabilities. If we think about the way that we have been stepped through the history of cryptography by exposure to so many of the techniques of the past is it not possible that the very concept of Kryptos is a nod towards its future? To be more specific Quantum Mechanics deals with the difficult philosophical questions that arise around the definition of reality and its relationship to measurement. The sculpture itself is inert, but as we've seen our observations of its mathematical properties breathe an imitation of our Solar System into its very being. When we stop looking then it returns to being an inert copper scroll latent with any possible meaning that anyone wishes to ascribe to it. To break Kryptos is to somehow extract information from it and pass information back in to the point where the observed and observer are reflected in some kind of symbiotic computation. Having read around the subject matter of Quantum Mechanics in the past I'm not sure I can find any better analogue of its mysteries than the one presented to us by Kryptos itself.